

Total No. of Questions : 8]

SEAT No. :

P812

[5870] - 1133

[Total No. of Pages : 2

T.E. (Computer Engineering)
DATA SCIENCE AND BIG DATA ANALYTICS
(2019 Pattern) (Semester - II) (310251)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 1) Neat diagrams must be drawn whenever necessary.
- 2) Figures to the right side indicate full marks.
- 3) Use of logarithmic tables slide rule, mollier charts, electronic pocket calculator and steam tables is allowed.
- 4) Assume suitable data, if necessary.

Q1) a) What is driving data deluge? Explain with one example. **[9]**

b) What is data science? Differentiate between Business Intelligence and Data Science. **[9]**

OR

Q2) a) What are the sources of Big Data. Explain model building phase with example. **[9]**

b) Explain big data analytics architecture with diagram. What is data discovery phase. Explain with example. **[9]**

Q3) a) Explain various data pre-processing steps. Discuss essential python libraries for preprocessing. **[8]**

b) What are association rules? Explain Apriori Algorithm in brief. **[9]**

OR

Q4) a) Explain the following

- i) Linear Regression
- ii) Logistic Regression **[8]**

b) Explain scikit-learn library for matplotlib with example. **[9]**

P.T.O.

- Q5) a)** Write short note on
- i) Time series Analysis
 - ii) TF - IDF. [9]

- b) What is clustering? With suitable example explain the steps involved in k - means algorithm. [9]

OR

- Q6) a)** Write short note on
- i) Confusion matrix
 - ii) AUC - ROC curve [9]

- b) Discuss Holdout method and Random Sub Sampling methods. [9]

- Q7) a)** With a suitable example explain Histogram and explain its usages. [8]

- b) Describe the Data visualization tool “Tableau”. Explain its applications in brief. [9]

OR

- Q8) a)** With a suitable example explain and draw a Box plot and explain its usages. [8]

- b) Describe the challenges of data visualization. Draw box plot and explain its usages. [9]

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